

CERN Summer Internship | June 2026

CMS Upgrade: Detector Laboratory Work

During my summer internship at CERN, I participated in laboratory activities associated with the CMS GEM detector upgrade. My work included assembly, sensor mounting and detector preparation. Handling detector components in a research environment gave me practical exposure to the precision, care and safety required when working on scientific instrumentation.





Cloud Chamber Kit: Making Particle Physics Accessible

I worked with a team to make particle physics more accessible to Indian school students through an affordable cloud chamber kit. Conventional cloud chambers use dry ice to reveal the tracks of charged particles, but sourcing, handling and cost can make them difficult for schools to use.

I investigated alternative cooling methods and took the lead in designing the kit's packaging. I explored a gel-pack-based cooling system as a possible alternative to dry ice. However, testing revealed that a conventional refrigerator could not achieve the required temperature, while keeping the gel pack sufficiently cold for sustained use was impractical. This helped me understand that a promising design must work under real-world constraints, not just in theory.

I also evaluated locally available components for a kit estimated to cost approximately ₹3,000 (USD 30). In designing its packaging, I focused on component organisation, safety guidance and ease of use for students and educators. I further helped students at BBLC construct a working cloud chamber, bringing our team's work into a hands-on learning environment.

